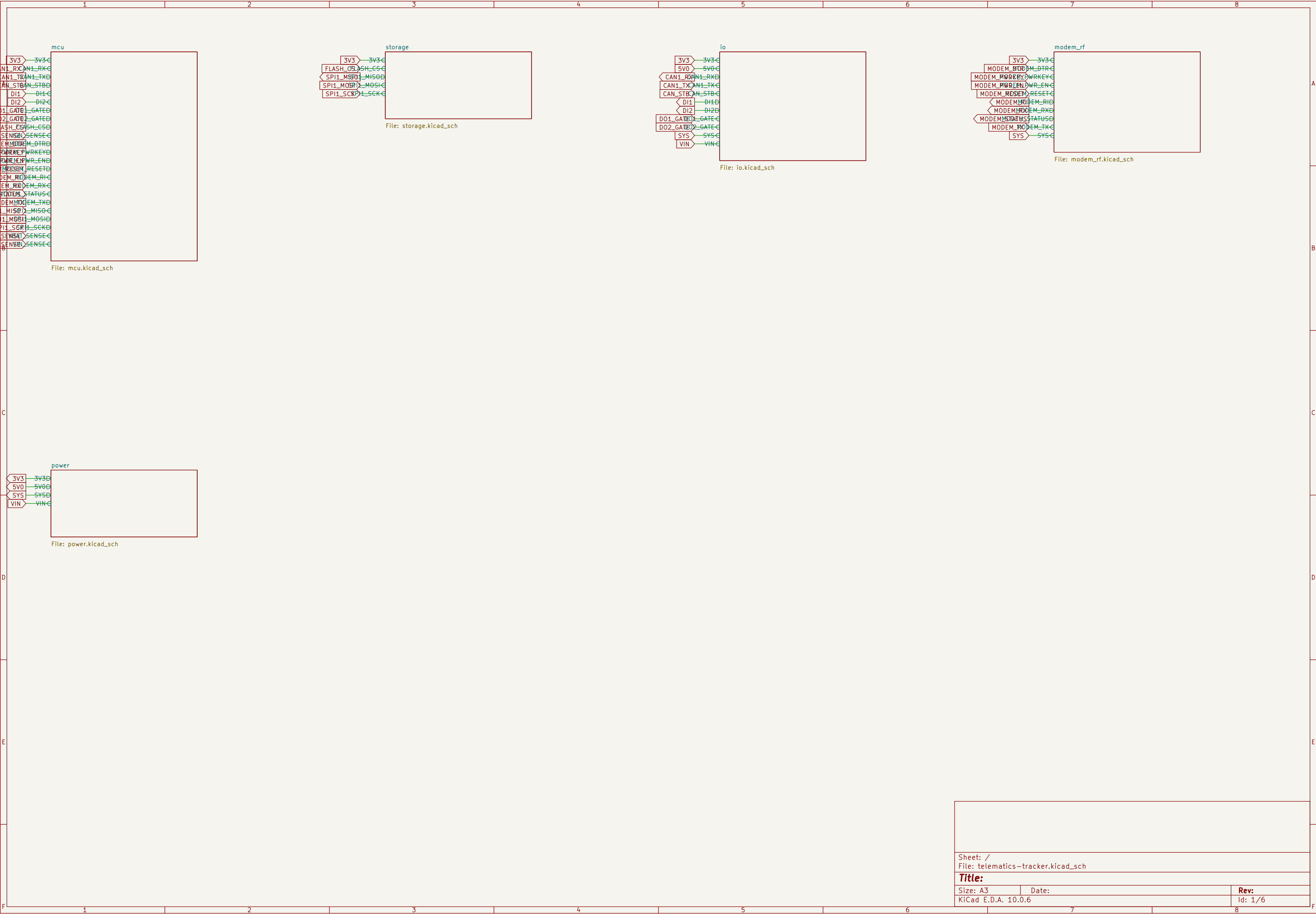
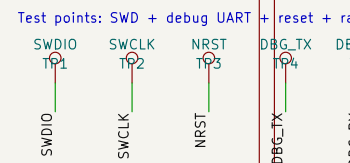


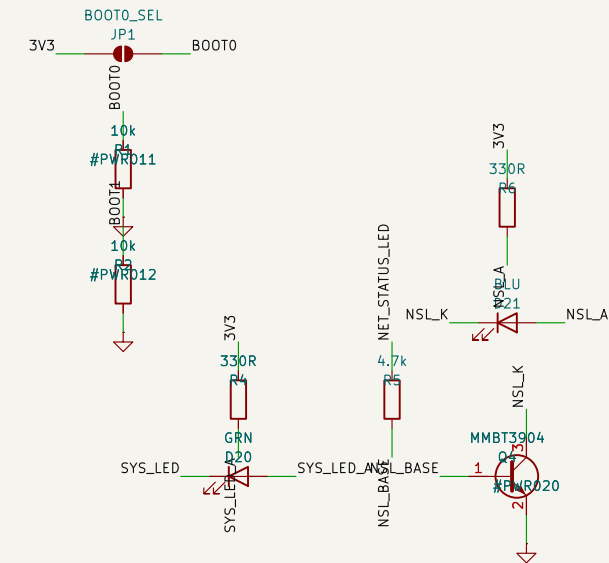


JTAG must be disabled in firmware (SWD retained) before PA15/PB3/PB4 are used as GPIO. CAN1 uses the PB8/PB9 remap.

RESV(10) tied high per datasheet (must NOT be GND); RESV-NC(11) floats.

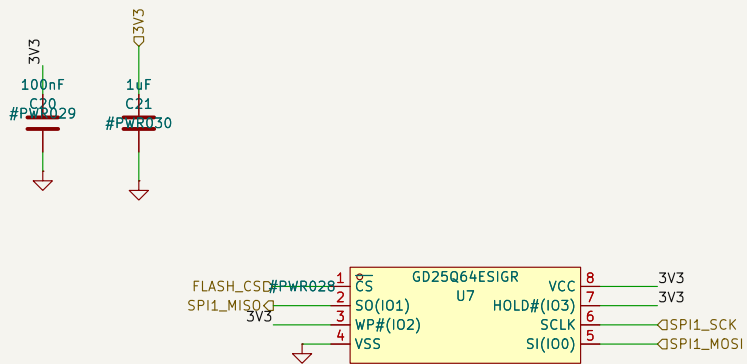


F-14 test points: modem UART pair, 3V3 (MCU) side of U8.



SPI NOR flash – telemetry ring buffer (≥ 6 days at 150 B/10 s).

U7 = GD25Q64ESIGR (tape & reel, C2831359) per F-3; W25Q64JVSSIQ (C179171/C2904572) is a verified pin- and command-compatible alternate.

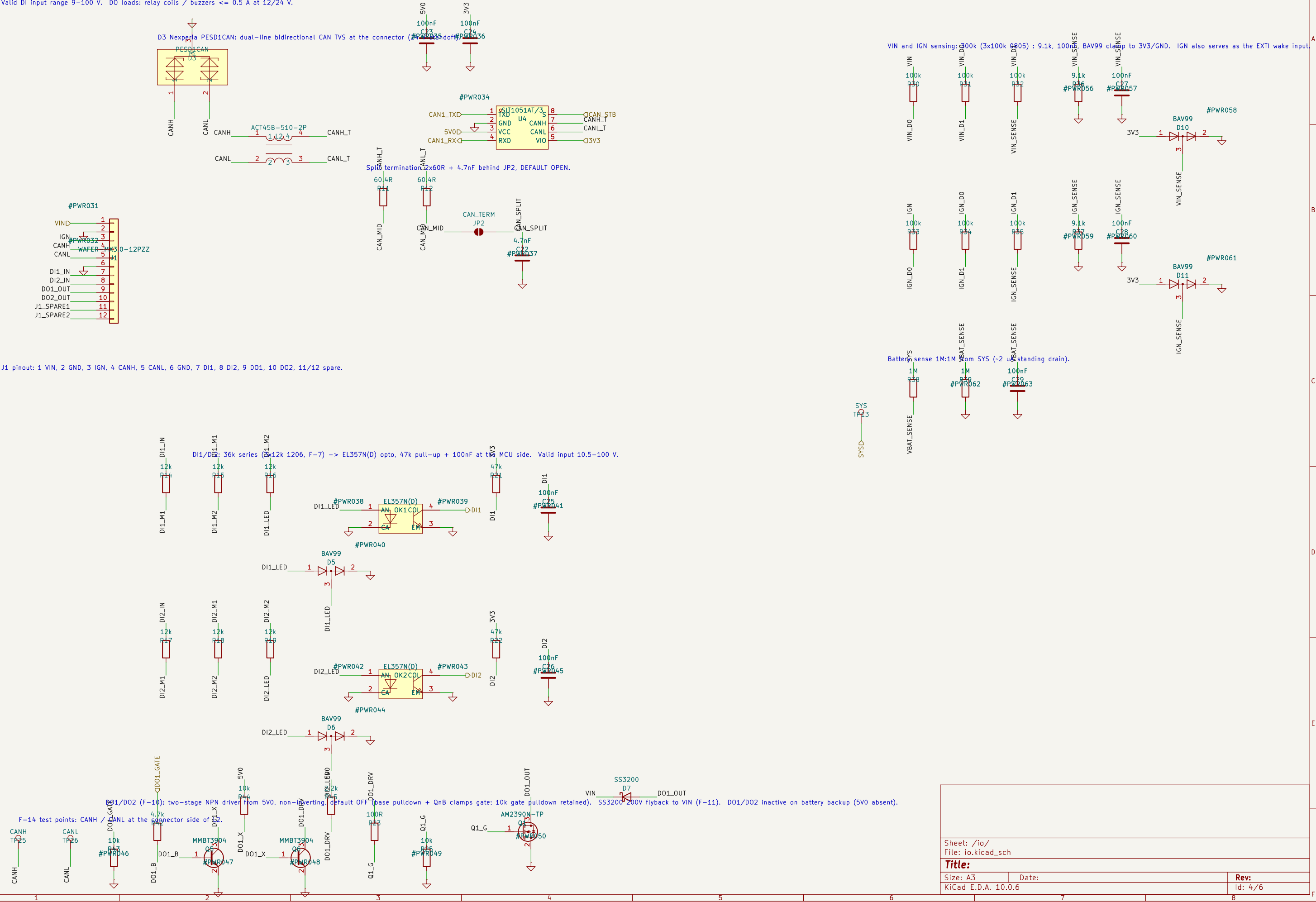


WP#(3) and HOLD#(7) tied to 3V3 per handoff section 5.

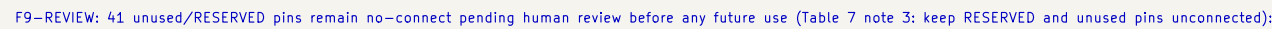
Sheet: /storage/ File: storage.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 3/6

MACHINE I/O – CAN, isolated digital inputs, low-side outputs, supply sensing. HV zone: J1 / dividers / DI series R / DO drains.

Valid DI input range 9–100 V. DO loads: relay coils / buzzers <= 0.5 A at 12/24 V.



F-9 SIGNED OFF 2026-09-04. All 43 GND pads are connected per Quectel EC200U HW Design V1.2 Table 7 p.21 / Table 9 p.36: "8, 9, 19, 22, 36, 46, 48, 50-54, 56, 72, 76, 85-112". F-9b reverse check clean - no GND pad omitted. Remaining NEEDS-HUMAN pins stay no-connect (Table 7 note 3).



THERMAL RELIEF PLAN (F-9, for layout): pads 85-112 are the central thermal/ground lattice – stitch every one straight down to the L2 solid GND plane with its own via (no thermal spokes, solid connection). Pad 76 is the local audio ground and pads 46/48/50/51 are the RF ground fence flanking ANT_GNSS(47)/ANT_MAIN(49) – fence pads get >=2 vias each, placed to keep the CPWG return path continuous. Perimeter grounds 8/9/19/22/36/52/53/54/56/72 get >=1 via each.

F-14 test point: VBAT_MODEM (probe transmit-burst rail sag).



POWER – 10.5–100 V front end, EG11752 buck (U5, approved with conditions a–c), BQ25606 charger, ME6211 3V3 LDO.

Front end: F1 (250 VDC interrupt) -> D1 S3M reverse block -> D2 SMDJ100A 3kW -> R80 1R 2512 anti-surge (F-15 approved) -> buck. Margin at 3.7A clamp is 34.9%. R80 at the 10.5V floor, full load $V_{FB}=9.16V$, $I_{FB}=4.4A$, 0.41W. 10R was a brown-out (no load-line solution) – see design-log F-15.

